

Civic Services Surveillance & Monitoring

Daily Progress Report — India-Specific Dataset Expansion

31 August 2026

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Goal for Today

Objective: find and import *genuinely India-specific* training data for each of the 7 civic-issue classes, organize it classwise, and separate a final curated "keep" set from the still-being-reviewed pool.

- Prior sessions' data was mostly non-Indian imagery (Roboflow/Kaggle sources picked for class relevance, not geography).
- Only 2–3 of ~13 existing sources were confirmed India-specific before today.
- Today: deep per-class research, strict visual verification of every candidate before use, classwise organization, and dashboard-based curation.

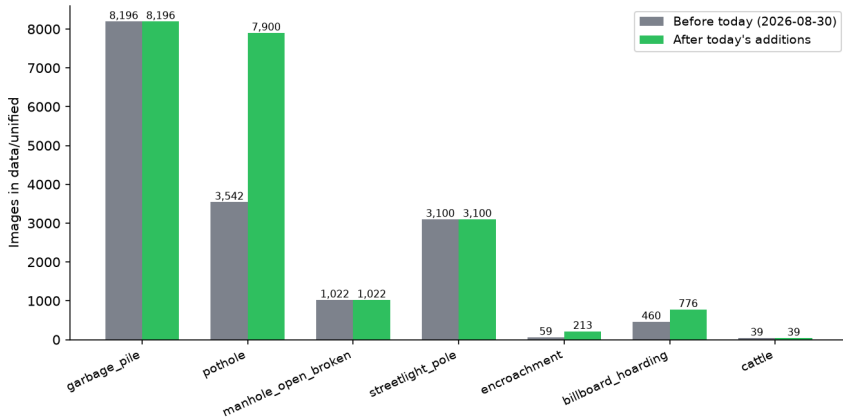
Method: Verify, Don't Trust the Title

Every candidate dataset was **downloaded and visually sample-checked** before being trusted — titles and workspace branding were repeatedly misleading:

- IIT-Madras/pothole-detection → sample image was an African township storefront.
- Smart-India-Hackathon/final-garbage → sample was a Bigstock stock photo of *Indonesian* grocery brands.
- sakshi-kumari/cow-indian → same dataset already rejected in an earlier session (tight face/nose crops, not road scenes) — re-surfaced by search without history cross-check.
- pib-e46kr/indian-bovine → genuine Indian breed dataset, but close-up farm/breed-showcase photography, not road scenes.

Lesson: accept nothing on title/workspace name alone — open real sample images first.

Data Collected Today — Summary

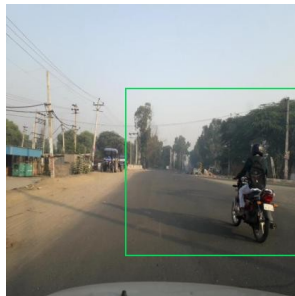


Data Collected — Source Detail

Class	Source	Added	Evidence of India origin
Pothole	indian-road-potholes (Roboflow)	4,306	Scooters, whitewashed compound walls, Indian apartment architecture
Pothole	bharat-pothole, India-tagged subset	52	Auto-rickshaw, Indian truck design, roadside signage
Billboard	advision/ooh-detection	316	360° footage on CG Road, Ahmedabad; AMTS (Ahmedabad transport) taxonomy
Encroachment	vendor/footpath-guide-2	154	Palm-lined streets, Indian utility poles, vendor cart on footpath
Streetlight	Chennai Working/NotWorking set	718*	Academic dataset (arXiv:2407.01117), captured across Chennai

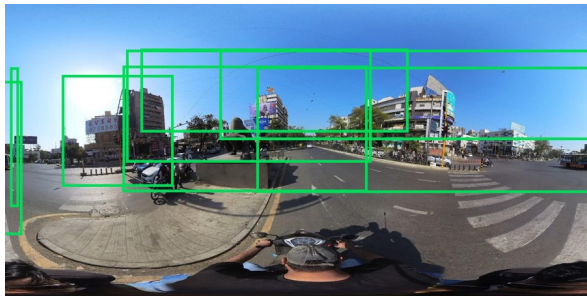
*Classification-format (whole-image label, no boxes) — see "Current Work in Progress".

Sample: Pothole (India-confirmed)



Left: indian-road-potholes. Right: bharat-pothole (India-tagged subset).

Sample: Billboard/Hoarding & Encroachment



Left: CG Road, Ahmedabad hoardings. Right: vendor cart obstructing footpath.

Sample: Streetlight — Working vs Not Working



Left: working. Right: non-functional. Chennai, India — 444 / 274 images.

Curated "Keep" Training Set Split Out

- Reviewed images in the dashboard (`review_tool.py`) marked **Keep** were moved (not copied) out of `data/unified/` into a new classwise folder: `data/training_data/<class>/{train,val}/{images,labels}`
- `data/unified/` now holds only the still-unreviewed / discarded pool — the dashboard's counts dropped accordingly.
- New read-only viewer built for the curated set:
`training_data_dashboard.py` — <http://127.0.0.1:5070>

Class	Curated (kept) images
manhole_open_broken	267
garbage_pile	128
pothole	125
streetlight_pole	101
billboard_hoarding	98
encroachment	59
cattle	39
Total	817

Streetlight functional/non-functional annotation (separate terminal session, in progress right now):

- The 718 Chennai streetlight photos are **classification-format** (whole-image Working/NotWorking label) — not usable directly by the box-based YOLO merge pipeline.
- First attempt: automatic zero-shot labeling (YOLO-World) — **rejected**, too inaccurate (~40% miss rate on real poles).
- Now building manual annotation instead: a dedicated tool (`streetlight_annotate.py`, port 5052) to draw boxes by hand.
- Status as of this report: tool is built and tested end-to-end, but **annotation has not started yet** — 0 / 443 Working, 0 / 273 NotWorking done.

This is the open item blocking streetlight_pole from benefiting from today's new data.

Classes Still Missing Good India Data

Despite a real search effort (Roboflow API, GitHub, Kaggle, academic sources), no credible India-specific data was found for:

- **Garbage on road** — the one "Smart India Hackathon" lead was actually an Indonesian stock photo; Kaggle lead unverifiable (no Kaggle credentials in this environment). **Weakest class along with cattle.**
- **Manhole (open/broken)** — one source has a "manhole" class, but doesn't distinguish open/broken from closed/intact — same contamination risk already flagged for the existing dataset, so not merged.
- **Cattle on road** — two more candidates checked, both close-up farm/breed photography, not road scenes. Existing 39-image India source remains the best (already strong, AP50 0.986 in the last trained model).

These three classes need either a Kaggle account + real verification, or self-collection — not more automated dataset search.

Next Steps

- 1 Finish manual streetlight Working/NotWorking annotation (other terminal, in progress).
- 2 Decide streetlight integration path: separate lit/unlit classifier vs. hand-boxed detection data.
- 3 Continue per-class review in `review_tool.py` for pothole and manhole_open_broken (largely untouched pools).
- 4 Set up Kaggle credentials to verify the one remaining garbage-dataset lead.
- 5 Retrain YOLO11n once the curated `data/training_data/` set is more complete — not yet retrained on any of today's changes.